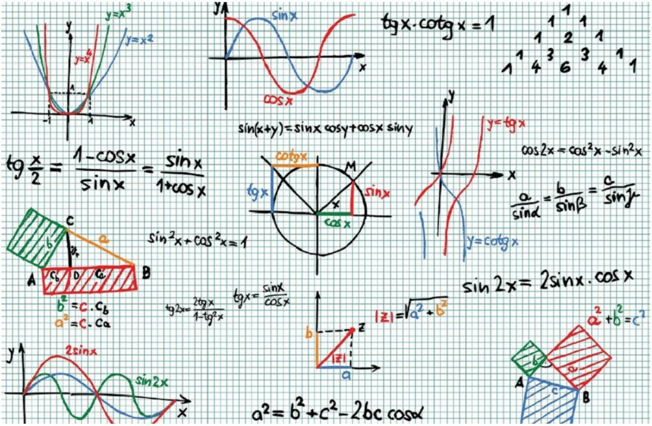
So What Are Figures, Graphs, and Plots in Octave?

This 12th article in the mathematical journey through open source covers mathematical visualisation in Octave.



Mathematics is incomplete without visualisation—without drawing the results and without plotting the graphs. Octave uses the powerful gnuplot as the back-end of its plotting functionality. And in the front-end it provides various plotting functions. Let's look at the most beautiful ones.

Basic 2D plotting

The most basic plotting is using the plot function, which takes the Cartesian x and y values. Additionally, you may pass the value as how to plot points or lines, their style, their colour, label, etc. Supported point styles are: +, *, o, x, ^, and lines are represented by -. Supported colours are: k (black), r (red), g (green), b (blue), y (yellow), m (magenta), c (cyan), and w (white). With this background, here is how you plot a sine curve:

```
$ octave -qf
octave:1> x = 0:0.1:2*pi;
octave:2> y = sin(x);
octave:3> plot(x, y, "^b");
octave:4> xlabel("x ->");
octave:5> ylabel("y = sin(x) ->");
octave:6> title(“Basic plot”);
octave:7>
```

xlabel() and *ylabel()* add the corresponding labels, and *title()* adds the title. Multiple plots can be done on the same axis as follows. Note the usage of *legend()* to mark the multiple plots.

```
$ octave -qf
octave:1> x = 0:0.1:2*pi;
octave:2> plot(x, sin(x), "*", x, 1 + sin(x), "-", x, cos(x), "o");
octave:3> xlabel("x ->");
octave:4> ylabel("y ->");
octave:5> legend("sine", "1 + sine", "cosine");
octave:6> title("Multiple plots");
octave:7>
```